Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/13/2006
PI Name:	Gundlach, Jens Ph.D.		
Project Title:	Preparation for space: torsion balances and equivalence principle tests for NASA		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	University of Washington		
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City:	Seattle	**State:**	WA
Zip Code:	98195-4290	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	04/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 09/30/2007 per PI (1/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	GSN 4335		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

Task Description:	Three closely interwoven research goals will be pursued: 1) Improving torsion balance techniques: Torsion balances have found multiple uses in NASA-related research. The sensitivity of torsion balances will be enhanced. 2) Conducting the most sensitive test of the weak equivalence principle: A test of the equivalence principle will be carried out using a rotating torsion balance. 3) Augmenting the test of the strong equivalence principle: NASA's Lunar laser ranging test of the strong equivalence principle remains ambiguous due to the different composition of the earth and the moon. A torsion balance measurement with test masses of similar composition as the earth and the moon will be carried out to remove this ambiguity. All three topics are of high relevance to present and future NASA endeavors. 1) Improving torsion balance techniques 2) Conducting a test of the weak equivalence principle 3) Augmenting the test of the strong equivalence principle new technologies fundamental science ground testing
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gravity gradient measurement and precision determination of local vertical.
Task Progress:	We have completed a measurement of the equivalence principle. The sensitivity of our measurement exeeds that of any previous measurement. We are in the process of writing the result up for publication. We will present upper limits for EP violations with ranges from 1m to 10Km and for 1000Km to infinity. Of particular interest will be our new limits on accelerations towards galactic dark matter. Numerous sources of systematic uncertainty were investigated and are well understood. Our next measurement will be directly relevant NASA's Laser Lunar Ranging effort.
Bibliography Type:	Description: (Last Updated: 07/28/2020)
Articles in Peer-reviewed Journals	Gundlach JH. "Laboratory tests of gravity." New Journal of Physics. 2005 Sep;7:205. http://dx.doi.org/10.1088/1367-2630/7/1/205 , Sep-2005
Articles in Peer-reviewed Journals	Schlamminger S, Choi K-Y, Wagner TA, Gundlach JH, Adelberger EG. "Test of the equivalence principle using a rotating torsion balance." Physical Review Letters. 2008 Feb;100(4):041101. https://doi.org/10.1103/PhysRevLett.100.041101 , Feb-2008